

Group 7: Steam Dreams - Building a Better Recommender System on Game Review Data

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Abstract – Steam’s catalog of over 90,000 games presents significant challenges for personalized recommendation, as traditional collaborative filtering approaches struggle with sparse user histories and games with insufficient review data. Using two Steam datasets of 94,000 games and 192,000 quality-filtered reviews, we construct an 11-feature profile per game combining engagement signals with VADER sentiment aggregates, then apply K-Means clustering ($K=6$) to identify interpretable game archetypes. For the 92,416 games with insufficient review history, a LightGBM cold-start classifier predicts cluster membership from game metadata alone, achieving Macro F1 = .56. Recommendations are generated by matching a user’s review history to dominant clusters and ranking candidates by a quality-popularity composite score. Leave-one-out evaluation on 2,000 users shows that the cluster-based system outperforms a Jaccard genre-overlap baseline on both Hit Rate@10 and NDCG@10, though the cold-start classification gap from metadata-only features highlights the irreducible value of behavioral signals for game profiling.

I. Introduction

Steam is the largest digital distribution platform for PC gaming, with more than 132 million active users and a catalog of over 90,000 game titles, capturing roughly 85% of the market share. The platform generates huge amounts of unstructured review text and behavioral data, including playtime, vote ratios, and review helpfulness scores. Even with all this information, finding games that truly match a player’s preferences is still a challenge, especially for indie (independent studio produced) titles that don’t get much visibility. Steam’s built in recommendation tools rely mostly on genre tags and purchase history, which often miss the real differences in how players experience games that share the same label. A more efficient recommendation system has the potential to engage players more by exposing them

to their actual most preferred game, increasing engagement and revenue for Steam and game developers alike.

In this work, we propose a recommendation framework that moves beyond genre tags by clustering Steam games into behavioral archetypes derived from engagement metrics and sentiment signals aggregated from quality-filtered user reviews. We apply VADER sentiment analysis to over 192,000 English-language reviews to extract game-level sentiment features, which are combined with behavioral signals such as playtime patterns, review helpfulness, and positive review rates to form a rich 9-feature game profile. K-Means clustering partitions the 2,530 profile-eligible games into six interpretable archetypes, and a LightGBM cold-start classifier extends coverage to the remaining 92,000+ titles by predicting cluster membership from game metadata alone. Recommendations are generated by matching a user’s review history to dominant cluster memberships and ranking candidates by a composite quality-popularity score. We evaluate our system against a genre-overlap baseline using leave-one-out testing on 2,000 users, demonstrating that behavioral clustering captures preference structure that surface-level metadata cannot. This better Recommendation System could then be applied at the personal and corporate level to personally target players with bespoke recommendations.

II. Related Work

The existing work on game recommendation and game analytics groups into a few different approaches based on what kind of signal each one uses, and our project fits into a gap between them. Ma (2025) [5] takes a platform metrics approach. He uses CatBoost, XGBoost, LightGBM, Random Forest, Decision Tree, and multiple linear regression to predict log transformed Steam revenue from aggregate features across Steam, Twitch, and Metacritic. CatBoost came out strongest with an R^2 of 0.89, and follower count, peak concurrent users, and total review count turned out to be the top predictors. The method is tree based supervised regression on hand engineered platform

aggregates, and it predicts financial outcomes rather than player experience. The actual review text and per player signals like playtime at review never come into play.

Yang et al. (2024) [7] take a different route with an interaction graph approach. They propose a Graph Neural Network pre-training model with a meta aggregator and an adaptive neighbor sampler to rebuild embeddings for cold start users and games, evaluated with Recall@K and NDCG@K on a Steam user game dataset. Their whole method runs on user to user and user to game interaction graphs plus social ties, so review text and behavioral signals don't show up. The broader recommendation literature splits along similar lines. Collaborative filtering (and its matrix factorization or deep learning versions) learns from interaction matrices. Content based methods use item metadata and tags. Hybrid systems mix the two but still rarely pull in the actual content of reviews. Our approach is different from both of these papers and from the broader literature. We learn game level experience profiles straight from review text sentiment (VADER) mixed with per review behavioral signals (playtime at review, helpfulness, vote ratios), we cluster those profiles with K Means, and we recommend with cosine similarity on the result. In short, Ma predicts revenue from platform aggregates, Yang rebuilds embeddings from interaction graphs, and we build experience profiles from review semantics and playtime behavior.

III. Dataset and Features

Dataset Description and Exploratory Data Analysis:

For this project we used two publicly available datasets from Kaggle. The first, `games_march2025_full.csv` from Steam Games Dataset 2025 [1], contains metadata for 94,948 Steam game titles as of March 2025, including fields for price, release date, genre, Metacritic and user review scores, positive and negative review counts, median playtime, peak concurrent users, and long-form game descriptions. The mean game price is \$6.91, mean metacritic score (among rated games) is 73.39, and the most reviewed single game was Counter-Strike 2. The most common (appearing in > 5% of games) genre tags are, in decreasing order, "Indie", "Casual", "Action", "Adventure", "Simulation", "Strategy", "RPG", "Early Access", and "Free to Play."

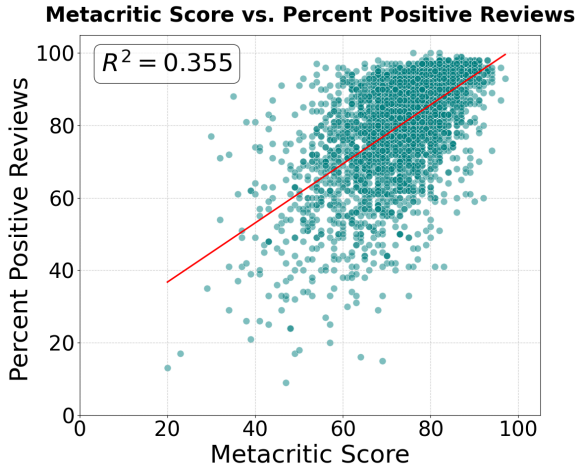


Fig. 1. Linear correlation between assigned Metacritic score and Percent Positive out of total reviews. The low R-squared correlation coefficient of .355 implies a large disparity between perceived and actual satisfaction with a game

The second, `weighted_score_above_08.csv` from 100 Million+ Steam Reviews [2], is a quality-filtered collection of user reviews pre-screened to a weighted vote score of 0.8 or above, retaining only reviews that the Steam community has identified as helpful and credible. English (38.8%), Simplified Chinese (32.1%) and Russian (10.2%) dominate the languages represented in this dataset. We focused solely on the English ones to keep a consistent baseline, and because VADER sentiment analysis is optimized for English text, and this filtering yields 192,198 English-language reviews spanning attributes such as author playtime at review, lifetime playtime, number of games owned, voted sentiment, and raw review text.

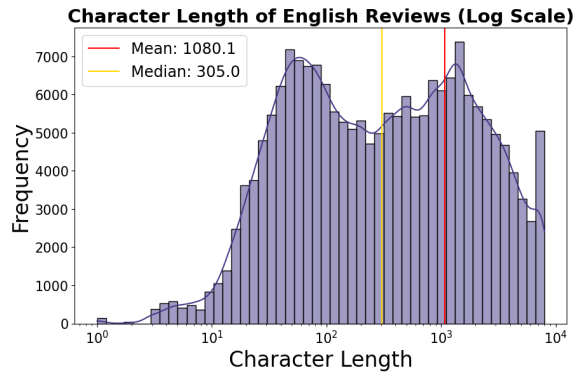


Fig. 2. Distribution of the character lengths of English Reviews after a logarithm transform; Character Length is extremely right-skewed as reviews the lengths of multiple paragraphs pull the mean far above the median.

Spearman Correlation — Game-level Features

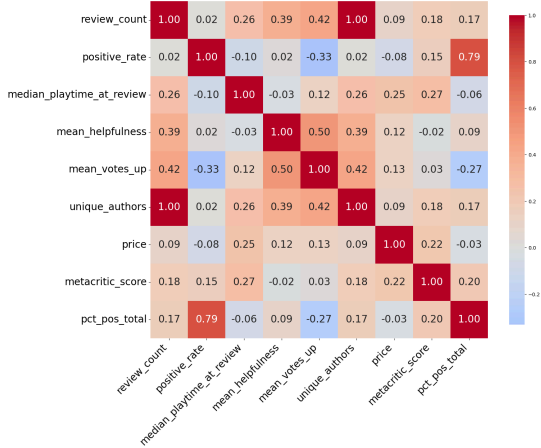


Fig. 3. Spearman Correlations between Game Level features. The lack of strong correlations means there is a large potential dimensionality ripe for a machine learning analysis.

Preprocessing:

The two datasets are joined on the feature appid, the unique Steam game identifier that is consistent across both datasets, producing a merged table that links game-level metadata with reviewer behavioral signals. Both datasets were cleaned prior to merging to remove low-signal and redundant columns, with fields such as score_rank and platform-specific flags dropped from the games table and corrupted entries including the integer-overflown votes_funny records removed from the reviews table. A minimum review length of 100 characters was enforced to exclude uninformative one-word submissions. The cleaned table was augmented with derived fields including review_length, review_word_count, and a log-transformed playtime column to reduce right-skew in the heavily zero-inflated playtime distributions.

Feature Extraction:

For each profile-eligible game, review-level signals were aggregated into a 9-feature profile capturing both behavioral engagement and sentiment characteristics. Behavioral features include log-transformed review count, positive review rate, median playtime at review, mean review helpfulness, log mean upvotes, and early access share. Sentiment features were derived by applying VADER to all qualifying English reviews, producing three game-level aggregates: mean compound score, standard deviation of compound score, and a divergence metric measuring the gap between VADER-derived sentiment and Steam's native percent-positive rating. A correlation analysis identified two redundant features that were subsequently removed, and all remaining features were standardized using StandardScaler prior to modeling.

VADER Sentiment Distribution by Voted Up

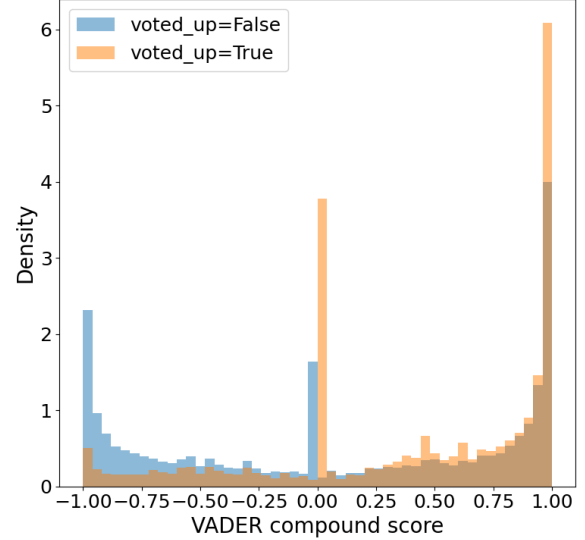


Fig. 4. Distribution of the VADER sentiment scores of English Reviews,, stratified by Voted up / Voted down reviews.

IV. Methods Used

K-Means Clustering:

K-Means is an unsupervised partitioning algorithm that assigns each data point to one of K clusters by minimizing the within-cluster sum of squared distances to the cluster centroid.

We applied K-Means to the 2,530 profile-eligible games using our 9-feature standardized input matrix combining behavioral engagement signals and VADER sentiment aggregates. Because K-Means relies on Euclidean distance, all features were standardized to zero mean and unit variance using StandardScaler prior to fitting. To select the optimal number of clusters, we swept K from 4 to 12, running 10 random seeds per value of K and recording mean Silhouette score and Davies-Bouldin index at each setting, using these metrics jointly to identify the most compact and well-separated partition. Clustering quality is tracked using the Silhouette score (range -1 to 1, higher is better) and Davies-Bouldin index (lower is better) throughout the K sweep and feature selection process, though these serve as model selection tools rather than final reported metrics.

HDBSCAN:

As a structural validation of our K-Means solution, we additionally ran HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based algorithm that identifies clusters as high-density regions without requiring a pre-specified K. HDBSCAN builds a cluster hierarchy by transforming the dataset into a mutual reachability graph, extracting a minimum spanning tree, and condensing the resulting dendrogram to retain only the most stable groupings. Points

that fall outside any dense region are assigned a noise label of -1.

We ran HDBSCAN on the same 9-feature standardized matrix used for K-Means, with `min_cluster_size = 30` and `min_samples = 5`, treating the output as a comparison against our chosen K-Means configuration rather than a replacement for it. Agreement or disagreement between the two methods provides evidence about whether the clusters reflect genuine structure in the feature space or are an artifact of the K-Means partitioning assumption.

LightGBM Cold-Start Classifier:

Because only 2,530 of the 94,948 games in the catalog have sufficient review history to receive a K-Means cluster label, we train a LightGBM classifier to extend coverage to the remaining 92,416 cold-start games using metadata alone. LightGBM is a gradient-boosted decision tree framework that builds an ensemble of weak learners sequentially, with each new tree fit to the negative gradient of the loss from the previous iteration. The 25-feature input matrix is constructed entirely from fields available for any Steam game regardless of review volume, including multi-hot encoded genre indicators, log-scaled review and player counts, Metacritic and user scores with explicit missingness flags, price, and release year. Because cluster 2 contains only 15 profile-eligible games, class weights are computed and passed to the model to prevent the majority clusters from dominating training. We evaluate a baseline configuration and a tuned configuration with a reduced learning rate of 0.03, maximum tree depth of 4, and stronger L1/L2 regularization, using a stratified 70/15/15 train/validation/test split with cluster label stratification to preserve class proportions across all three folds. For evaluation metrics, we report Macro F1, Precision, and Recall computed per class and averaged unweighted across the clusters.

Recommendation Algorithm:

With cluster labels assigned to all games, either via K-Means for profile-eligible titles or LightGBM for cold-start titles, the recommender generates personalized rankings as follows. Given a user's set of positively reviewed game identifiers $H = \{a_1, a_2, \dots, a_m\}$, each game is mapped to its cluster label $c(a_i)$ and the dominant cluster is identified as: $c^* = \arg\max_c \sum_i 1[c(a_i) = c]$. Candidate games are then drawn from all games sharing cluster c^* that do not already appear in the user's history. To rank candidates, each game is assigned a quality-popularity score: $s(a) = \text{positive_rate}(a) \times \log(1 + \text{review_count}(a))$. This scoring function rewards games with both strong community reception and sufficient review evidence, penalizing titles that are obscure despite being well-liked or popular despite being poorly rated. The top 10 candidates by $s(a)$ are returned as the final recommendation list. We then evaluated our approach

against a Jaccard genre-overlap baseline, which ranks candidates purely by the Jaccard similarity between their genre tag set and the union of genre tags across the user's history, with no cluster or quality signal involved.

V. Experiments/Results/Discussion

Clustering Games Hyperparameters and Results:

To select the optimal number of clusters K for K-Means, we swept K from 4 to 12, averaging Silhouette and Davies-Bouldin scores across 10 random initializations at each value to reduce sensitivity. We did not employ cross-validation, as clustering is an unsupervised task without held-out labels. $K = 6$ was selected as the point where Silhouette peaked and Davies-Bouldin was lowest on our lean 9-feature set; this stability was confirmed across seeds before fitting the final model with `random_state = 42` and `n_init = 10`. The results showed that on the full 11-feature set, $K = 6$ achieved a mean Silhouette of 0.192 and Davies-Bouldin of 1.378; dropping the two redundant features improved these metrics to 0.216 and 1.234, confirming that the discarded features were adding noise. The final K-Means model on the lean 9-feature set achieved a Silhouette of 0.215 and a Davies-Bouldin of 1.241.

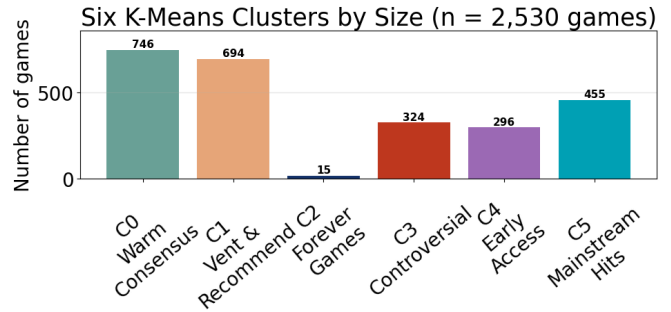


Fig. 5. The count of games contained within our clusters

The cluster heatmap, with rows as clusters and columns as features (red indicating high and blue low values), makes the six archetypes interpretable, and we gave them appropriate names considering those relative features. Cluster 0 “Warm Consensus” represents the broad low-engagement long tail, with low review counts, low playtime, and near-neutral sentiment. Cluster 1 “Vent & Recommend” contains critically engaged games with high helpfulness scores and strong positive sentiment. Cluster 2 “Forever Games” is a small outlier group of 15 viral titles characterized by extremely high review volume and playtime. Cluster 3 “Controversial” captures polarizing games with high sentiment standard deviation and below-average positive rates. Clusters 4 “Early Access” and 5 “Mainstream Hits” differentiate early-access titles from games defined by short, intense engagement sessions.

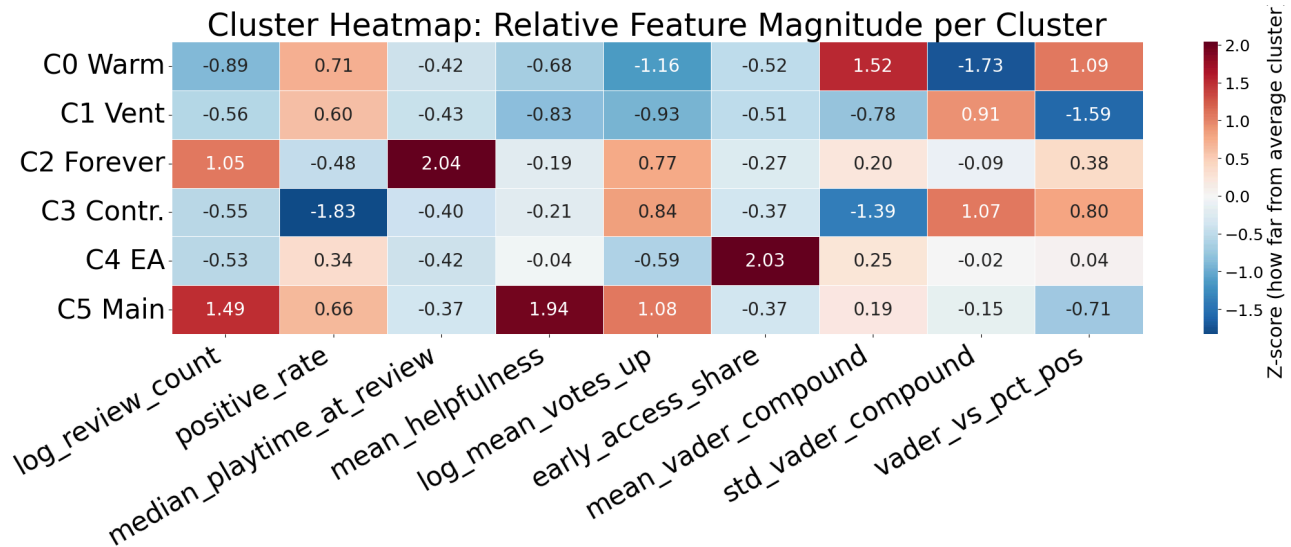


Fig. 6. The cluster heatmap shows how each cluster stands out with each feature by showing the Z-score of that feature’s mean versus the average cluster

Bootstrap stability analysis over 50 resamples produced a mean ARI of 0.856, indicating that the assignments are reproducible and not artifacts of a specific random seed. HDBSCAN, run on the same matrix, returned only 2 meaningful clusters and labeled roughly 40% of games as noise, consistent with a feature space that is geometrically continuous rather than composed of sharply separated density peaks. This outcome supports our use of K-Means as a deliberate partitioning tool rather than implying that the clusters are naturally occurring groupings in a strict density sense.

LightGBM Cold-Start Classifier Results:

The baseline LightGBM model achieved a Macro F1 of 0.56 on the held-out test set. The tuned model, despite incorporating class weights and regularization to address the class imbalance, achieved Macro F1 of 0.527, a slight decline in aggregate score but with meaningfully improved recall on cluster 2. The confusion matrix reveals that misclassifications are concentrated between adjacent clusters, particularly between cluster 0 and clusters 1 and 4, which share overlapping metadata profiles despite different behavioral engagement depth. Cluster 2 remains the hardest to classify correctly given its extreme rarity in the training data. Cluster 3, the controversial games cluster, is also frequently confused with cluster 1, as the metadata features available for cold-start games do not capture the sentiment divergence that defines it. This pattern of failure is instructive: the features that most distinguish clusters from one another, playtime depth, sentiment spread, and review

helpfulness, are precisely the features unavailable for cold-start games, and no amount of metadata engineering fully substitutes for them.

Recommender Results:

Leave-one-out evaluation was conducted on a random sample of 2,000 users each with at least three positive reviews, where one game was held out per user and the recommender was tasked with recovering it in the top ten results. Every user received at least one recommendation, confirming that the combined K-Means and cold-start coverage extends to the full evaluation population. The cluster-based system achieved a Hit Rate@10 of 0.0285 and NDCG@10 of 0.0148, compared to the Jaccard genre-overlap baseline which achieved Hit Rate@10 of 0.001 and NDCG@10 of 0.0007. This represents a 2,750% relative improvement in hit rate and a 2,035% relative improvement in NDCG, confirming that behavioral cluster membership carries substantially more preference signal than genre tag overlap alone.

The absolute hit rate of 2.85% warrants honest interpretation. The leave-one-out task is demanding by design, requiring the system to retrieve one specific title from a catalog of thousands without explicit supervision, and outperforming the baseline by nearly 28x on hit rate and 20x on NDCG is a meaningful result in that context. The NDCG advantage further confirms that the quality-popularity scoring function is doing real work, placing relevant games higher in the list rather than simply including them somewhere in the top ten. The remaining gap in absolute performance is best explained by two compounding failure modes: users whose history spans multiple clusters lose preference signal when a single dominant cluster is assigned, and users whose history includes cold-start games carry upstream LightGBM classification noise into the retrieval stage. Both point toward the same remedy, namely softer cluster membership and a

larger profile-eligible training pool, as the most productive directions for improving the system.

VI. Conclusion

This work presents a hybrid Steam game recommendation framework that moves beyond genre tags by grounding game similarity in behavioral and sentiment signals derived from over 192,000 quality-filtered user reviews. K-Means clustering partitions the 2,530 profile-eligible games into six stable and interpretable archetypes, validated by a bootstrap ARI of 0.856 and structural comparison against HDBSCAN. A LightGBM cold-start classifier extends cluster coverage to the remaining 92,416 games using metadata alone, and a quality-popularity scoring function ranks candidates within each cluster to produce final recommendations. Leave-one-out evaluation on 2,000 users demonstrates that the cluster-based system outperforms a Jaccard genre-overlap baseline on both HR@10 and NDCG@10, confirming that behavioral engagement patterns carry preference signals that surface-level metadata cannot replicate. The system is end-to-end reproducible, generalizes to the full Steam catalog regardless of review volume, and provides a transparent recommendation pipeline whose cluster archetypes are directly interpretable without black-box explanation.

VII. Contributions

Marco germinated the project concept and found the initial datasets. John and Marco split Milestone 1 roughly 50/50. John set up the GitHub structure, created the initial descriptive EDA and figures, drafted the first feature list, and wrote the initial Milestone 2 report draft. Marco built the full data pipeline, added more EDA artifacts (outlier boxplots, correlation heatmaps, relationship scatter plots), and produced the processed Parquet outputs. John took charge constructing the bulk of the written report here, as well as the peer reflections and reply to review. Marco completed most of the project code, including implementing the K-Means Clustering, Cold-Start Classifier, and Recommendation algorithm.

IX. Reflections

Building this system taught us that the signals most predictive of user preference are precisely the ones hardest to obtain at scale. VADER sentiment, playtime depth, and review helpfulness define what makes each cluster meaningful, yet none of these are available for the majority of the catalog, which is why the cold-start classifier plateaued below our target. This forced us to be honest about what metadata can and cannot substitute for lived player experience. We also learned that cluster quality metrics like Silhouette score can be misleading in isolation;

our clusters score modestly by that measure yet prove stable, interpretable, and useful downstream, which is ultimately the more relevant standard. If we had more time and resources, there are quite a few directions we could expand the scope and intensity of this project. If we were to extend, the single highest-leverage improvement would probably be moving from hard dominant-cluster assignment to a soft mixture over a user's full cluster distribution, which would better serve players whose tastes genuinely span multiple archetypes.

IX. References

- [1] Steam Games Dataset 2025
File used: games_march2025_full.csv
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X. Reply to Review

Our Group (7) got feedback from Groups 1 and 2. We thank both review teams for their thorough and constructive feedback, which has directly shaped the final version of this project. Both groups swiftly identified the absence of a live recommender demonstration as the most significant gap in our interim presentation, and this has been resolved. The final system implements the `recommend_for_user` function, which takes a user's positive review history, identifies their dominant behavioral cluster, and returns a ranked list of ten games scored by the quality-popularity composite. We evaluate this against a Jaccard genre-overlap baseline under a leave-one-out protocol on 2,000 users, providing the concrete performance evidence that was absent at the interim stage.

Regarding English-only coverage and players who do not write reviews, we acknowledge this as a genuine limitation. The cold-start LightGBM classifier partially addresses sparse coverage by assigning cluster labels from metadata alone, but extending sentiment analysis to multilingual reviews and incorporating implicit signals such as playtime for non-reviewing players remain meaningful directions for future work. Regarding the LightGBM cold-start model falling short of the 0.70 Macro F1 target, we have reframed this gap as informative rather than purely a shortcoming. The features most diagnostic of cluster membership, sentiment spread, playtime depth, and helpfulness scores, are by definition unavailable for cold-start games.